BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Bubble Sort and Insertion Sort

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Define linear search.

2. Where is index 0 in a one-dimensional array?

3. Does linear search require sorted data?

4. A list is [4, 2, 9, 1]. How many comparisons are made to find 9 with linear search?

Today's objectives

Explain how bubble sort compares neighbouring values. Describe insertion sort's growing sorted section.

Key vocabulary - Read, cover, recall

bubble sort
Repeatedly compares and swaps adjacent out-of-order values.

insertion sort
Inserts each next item into a growing sorted section.

pass
One traversal of adjacent values in a sort.

key
Current value being inserted into a sorted section.

Write one definition in your own words:

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Learn and practise

New learning - 2.1.3

Bubble sort	Compare adjacent values and swap those in the wrong order; repeat passes until no swaps are needed.
After each pass	For ascending bubble sort, the largest remaining value moves towards the end on each full pass.
Insertion sort	Treat the first item as a sorted section; take the next key item and insert it into the correct place.
Keep a trace	Show each swap or insertion rather than jumping straight to the final sorted result.

Bubble: one pass

[5,1,4,2] -> [1,5,4,2] -> [1,4,5,2] -> [1,4,2,5]

Insertion: grow the sorted section

[6,2,5] -> [2,6,5] -> [2,5,6]

Show intermediate swaps or insertions, not just the result.

I Do - Worked example

Show one full bubble-sort pass through [5, 1, 4, 2].

1. Compare 5 and 1: swap · [1, 5, 4, 2].
2. Compare 5 and 4: swap · [1, 4, 5, 2].
3. Compare 5 and 2: swap · [1, 4, 2, 5].
4. End of first pass: 5 has reached its final position.
5. More passes are needed to sort the remaining values.

We Do - Guided practice

Perform one bubble-sort pass on [4, 2, 3].

Use the model. Write your answer in BLACK PEN:

BLACK PEN Answer all questions first. **GREEN PEN** Then tick and correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Read the code before you write or test anything.

```
a = 4
b = 2
if a > b then
  temp = a
  a = b
  b = temp
endif
print(a)
```

Trace one neighbouring swap of two file sizes (in GB).

1. Predict the printed file size.

2. Change a to 1.

3. Explain how comparing file sizes helps calculate total storage requirements.

You Do - Independent exam questions

Q1. State what bubble sort compares. [1 mark]

Q2. Show one bubble-sort pass on [3, 1, 2], giving the array after each swap. [3 marks]

Q3. Explain how insertion sort orders the list [4, 1, 3]. [4 marks]

Exit Ticket - True or False?

Answer each statement before turning to the answers.

1. One bubble-sort pass always completely sorts any list. ☐ T ☐ F

2. Insertion sort grows a sorted section. ☐ T ☐ F

3. Bubble sort compares neighbouring items. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick what is correct, then use the model to fix mistakes.
Do not just copy answers; explain what you changed.

Do Now - Answers

Q1 Checking values in turn until target is found or the list ends. ☐

Q2 At the first element. ☐

Q3 No. ☐

Q4 Three comparisons: 4, then 2, then 9. ☐

Code First - Model

Initially values 4 and 2 swap, so a becomes 2. Changing the first a to 1 means no swap, so a prints 1. Comparing file sizes helps order files; total storage requires adding all their sizes, so 4 GB and 2 GB together require 6 GB.

We Do - Model answer

Swap 4/2 · [2, 4, 3]; swap 4/3 · [2, 3, 4].

Exit Ticket - Check

1. FALSE - False: revise the new learning.
2. TRUE - True: the next value is inserted into the correct place.
3. TRUE - True: adjacent values are compared and may be swapped.

You Do - Self-mark your exam answers

Q1 Adjacent (neighbouring) values [1]. ☐

Q2 [1, 3, 2] [1]; [1, 2, 3] [1]; largest item reaches end [1]. ☐

Q3 First item 4 is initial sorted section [1]; insert 1 before 4 · [1, 4, 3] [1]; take key 3 [1]; insert between 1 and 4 · [1, 3, 4] [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW complete answer. Show working: